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Modul 08

queue.h

```
1  #ifndef QUEUE_H_INCLUDED
2  #define QUEUE_H_INCLUDED
3  #include <iostream>
4
5  using namespace std;
6
7  typedef struct elmQueue *address;
8
9  struct infotype {
10     string nama;
11     int usia;
12     string pekerjaan;
13     bool prioritas;
14     int nomorAntrean;
15 };
16
17 struct Queue {
18     address head;
19     address tail;
20 };
21
22 struct elmQueue {
23     infotype info;
24     address next;
25 };
26
27 void createQueue_103032400084(Queue &Q);
28 bool isEmpty_103032400084(Queue Q);
29 address allocate_103032400084(string nama, int usia, string pekerjaan, int nomorAntrean);
30 address front_103032400084(Queue Q);
31 address back_103032400084(Queue Q);
32 void enqueue_103032400084(Queue &Q, address p);
33 void dequeue_103032400084(Queue &Q, address &p);
34 int size_103032400084(Queue Q);
35 void printPatient_103032400084(address p);
36 void serveQueue_103032400084(Queue &Q);
37 void reassignQueue_103032400084(Queue &Q);
38 void emergencyHandle_103032400084(Queue &Q, int nomorAntrean);
39
40 #endif // QUEUE_H_INCLUDED
41
```

queue.cpp

```

1  #include "queue.h"
2
3  void createQueue_103032400084(Queue &Q) {
4      Q.head = nullptr;
5      Q.tail = nullptr;
6  }
7
8  bool isEmpty_103032400084(Queue Q) {
9      return (Q.head == nullptr && Q.tail == nullptr);
10 }
11
12 address allocate_103032400084(string nama, int usia, string pekerjaan, int nomorAntrean) {
13     address p = new elmQueue;
14     infotype x;
15
16     x.nama = nama;
17     x.usia = usia;
18     x.pekerjaan = pekerjaan;
19     x.prioritas = usia >= 60 || pekerjaan == "tenaga_kesehatan";
20     x.nomorAntrean = nomorAntrean;
21     p -> info = x;
22     p -> next = nullptr;
23
24     return p;
25 }
26
27 address front_103032400084(Queue Q) {
28     return Q.head;
29 }
30

```

```

31 address back_103032400084(Queue Q) {
32     return Q.tail;
33 }
34
35 void enqueue_103032400084(Queue &Q, address p) {
36     address temp;
37     bool inputPasienBukanPrioritas, semuaPasienPrioritas, tidakAdaPasienPrioritas;
38
39     if (!isEmpty_103032400084(Q)) {
40         inputPasienBukanPrioritas = !p -> info.prioritas;
41         semuaPasienPrioritas = Q.tail -> info.prioritas;
42         tidakAdaPasienPrioritas = !Q.head -> info.prioritas;
43     }
44
45     if (isEmpty_103032400084(Q)) {
46         Q.head = p;
47         Q.tail = p;
48     } else if (inputPasienBukanPrioritas || semuaPasienPrioritas) {
49         Q.tail -> next = p;
50         Q.tail = p;
51     } else if (tidakAdaPasienPrioritas) {
52         p -> next = Q.head;
53         Q.head = p;
54     } else {
55         temp = Q.head;
56
57         while (temp -> next != nullptr && temp -> next -> info.prioritas) {
58             temp = temp -> next;
59         }
60

```

```

61         p -> next = temp -> next;
62         temp -> next = p;
63     }
64 }
65
66 void dequeue_103032400084(Queue &Q, address &p) {
67     if (!isEmpty_103032400084(Q)) {
68         p = Q.head;
69         Q.head = p -> next;
70
71         if (Q.head == nullptr) {
72             Q.tail = nullptr;
73         }
74     } else {
75         p = nullptr;
76     }
77 }
78
79 int size_103032400084(Queue Q) {
80     int count;
81     address p;
82
83     count = 0;
84     p = Q.head;
85
86     while (p != nullptr) {
87         count++;
88         p = p -> next;
89     }
90

```

```

91     return count;
92 }
93
94 void printPatient_103032400084(address p) {
95     if (p != nullptr) {
96         cout << "-----" << endl;
97         cout << "nama: " << p -> info.nama << endl;
98         cout << "usia: " << p -> info.usia << endl;
99         cout << "pekerjaan: " << p -> info.pekerjaan << endl;
100        cout << "prioritas: " << p -> info.prioritas << endl;
101        cout << "nomor antrean: " << p -> info.nomorAntrean << endl;
102    }
103 }
104
105 void serveQueue_103032400084(Queue &Q) {
106     address p;
107     int i = 0;
108
109     while (i < 5 && !isEmpty_103032400084(Q)) {
110         dequeue_103032400084(Q, p);
111         printPatient_103032400084(p);
112         cout << "Vaksinasi berhasil." << endl;
113         i++;
114     }
115 }
116
117 void reassignQueue_103032400084(Queue &Q) {
118     address p;
119     p = Q.head;
120

```

```

121     while (p != nullptr) {
122         if (p -> info.prioritas == false) {
123             p -> info.prioritas = true;
124         }
125         p = p -> next;
126     }
127     cout << "Data prioritas pasien telah di update" << endl;
128 }
129
130 void emergencyHandle_103032400084(Queue &Q, int nomorAntrean) {
131     address p, temp;
132     bool ketemu;
133
134     if (!isEmpty_103032400084(Q)) {
135         if (Q.head -> info.nomorAntrean == nomorAntrean) {
136             Q.head -> info.prioritas = true;
137         } else {
138             temp = Q.head;
139             p = Q.head -> next;
140             ketemu = false;
141
142             while (p != nullptr && !ketemu) {
143                 if (p -> info.nomorAntrean == nomorAntrean) {
144                     ketemu = true;
145                 } else {
146                     temp = p;
147                     p = p -> next;
148                 }
149             }
150
151             if (ketemu) {
152                 temp -> next = p -> next;
153                 if (p -> next == nullptr) {
154                     Q.tail = temp;
155                 }
156                 p -> info.prioritas = true;
157                 p -> next = Q.head;
158                 Q.head = p;
159             }
160         }
161     }
162 }
163

```

main.cpp

```

1 | #include "queue.h"
2 |
3 | int main() {
4 |     Queue Q;
5 |     address p;
6 |     int usia, inputAntrean, banyakPasien, inputPrioritas;
7 |     string nama, pekerjaan;
8 |
9 |     createQueue_103032400084(Q);
10 |    cout << "masukkan banyak pasien yang ingin di input: ";
11 |    cin >> banyakPasien;
12 |    cout << endl;
13 |
14 |    for (int i = 0; i < banyakPasien; i++) {
15 |        cout << "Pasien ke-" << i + 1 << endl;
16 |        cout << "nama: ";
17 |        cin >> nama;
18 |
19 |        cout << "usia: ";
20 |        cin >> usia;
21 |
22 |        cout << "pekerjaan: ";
23 |        cin >> pekerjaan;
24 |
25 |        cout << "nomor antrean: ";
26 |        cin >> inputAntrean;
27 |
28 |        p = allocate_103032400084(nama, usia, pekerjaan, inputAntrean);
29 |        enqueue_103032400084(Q, p);
30 |        cout << endl;
31 |    }
32 |
33 |    cout << endl;
34 |    cout << "banyak pasien dalam antrean: " << size_103032400084(Q) << endl;
35 |
36 |    cout << "serve queue: " << endl;
37 |    serveQueue_103032400084(Q);
38 |    cout << endl;
39 |
40 |    cout << "emergency handle dan serve queue: " << endl;
41 |    cout << "masukkan nomor antrean yang ingin di prioritaskan: ";
42 |    cin >> inputPrioritas;
43 |    emergencyHandle_103032400084(Q, inputPrioritas);
44 |    serveQueue_103032400084(Q);
45 |    cout << endl;
46 |
47 |    cout << "reassing queue dan serve queue: " << endl;
48 |    reassignQueue_103032400084(Q);
49 |    serveQueue_103032400084(Q);
50 |
51 |    return 0;
52 | }
53 |

```

output

1.4.1 enqueue

masukkan banyak pasien yang ingin di input: 11

Pasien ke-1

nama: andi

usia: 25

pekerjaan: guru

nomor antrean: 1

Pasien ke-2

nama: bela

usia: 42

pekerjaan: tenaga_kesehatan

nomor antrean: 2

Pasien ke-3

nama: citra

usia: 33

pekerjaan: mahasiswa

nomor antrean: 3

Pasien ke-4

nama: doni

usia: 45

pekerjaan: petani

nomor antrean: 4

Pasien ke-5

nama: eka

usia: 70

pekerjaan: pensiunan

nomor antrean: 5

Pasien ke-6

nama: faisal

usia: 29

pekerjaan: karyawan_swasta

nomor antrean: 6

Pasien ke-7

nama: gina

usia: 54

pekerjaan: pedagang

nomor antrean: 7

Pasien ke-8

nama: heri

usia: 66

pekerjaan: tenaga_kesehatan

nomor antrean: 8

```
Pasien ke-9  
nama: indah  
usia: 31  
pekerjaan: wirausaha  
nomor antrean: 9  
  
Pasien ke-10  
nama: joni  
usia: 40  
pekerjaan: supir  
nomor antrean: 10  
  
Pasien ke-11  
nama: kiki  
usia: 58  
pekerjaan: ibu_rumah_tangga  
nomor antrean: 11
```

1.4.2 size

```
banyak pasien dalam antrean: 11
```

1.4.3 serveQueue

```
serve queue:
-----
nama: bela
usia: 42
pekerjaan: tenaga_kesehatan
prioritas: 1
nomor antrean: 2
Vaksinasi berhasil.
-----
nama: eka
usia: 70
pekerjaan: pensiunan
prioritas: 1
nomor antrean: 5
Vaksinasi berhasil.
-----
nama: heri
usia: 66
pekerjaan: tenaga_kesehatan
prioritas: 1
nomor antrean: 8
Vaksinasi berhasil.
-----
nama: andi
usia: 25
pekerjaan: guru
prioritas: 0
nomor antrean: 1
Vaksinasi berhasil.
-----
nama: citra
usia: 33
pekerjaan: mahasiswa
prioritas: 0
nomor antrean: 3
Vaksinasi berhasil.
```


1.4.4 emergencyHandle

```
emergency handle dan serve queue:
masukkan nomor antrean yang ingin di prioritaskan: 4
-----
nama: doni
usia: 45
pekerjaan: petani
prioritas: 1
nomor antrean: 4
Vaksinasi berhasil.
-----
nama: faisal
usia: 29
pekerjaan: karyawan_swasta
prioritas: 0
nomor antrean: 6
Vaksinasi berhasil.
-----
nama: gina
usia: 54
pekerjaan: pedagang
prioritas: 0
nomor antrean: 7
Vaksinasi berhasil.
-----
nama: indah
usia: 31
pekerjaan: wirausaha
prioritas: 0
nomor antrean: 9
Vaksinasi berhasil.
-----
nama: joni
usia: 40
pekerjaan: supir
prioritas: 0
nomor antrean: 10
Vaksinasi berhasil.
```

1.4.5 reassignQueue

```
reassign queue dan serve queue:  
Data prioritas pasien telah di update  
-----  
nama: kiki  
usia: 58  
pekerjaan: ibu_rumah_tangga  
prioritas: 1  
nomor antrean: 11  
Vaksinasi berhasil.
```